

Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 6: Adaptations of the Human Digestive System — Structure Meets Function

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	A	C	C	D	B

USE AFTER: OBSERVE PHASE — Observe: Evidence Table using the labelled villus cross-section (resource pack item A)

Q1. A Grade 10 learner studies a diagram of one villus in the small intestine. Inside the villus is a network of tiny blood capillaries very close to the surface cells. What is the main advantage of this capillary network?

- A. It makes mucus to protect the villus from enzymes
- B. It carries absorbed nutrients such as glucose into the bloodstream**
- C. It pushes food along the small intestine
- D. It produces amylase to digest the starch from ugali

Correct answer: B. The capillary network inside each villus lets glucose and other nutrients that cross the epithelial cells enter the bloodstream straight away. Mucus comes from goblet cells, and enzymes are not made by capillaries.

USE AFTER: MODEL BUILDING PHASE — Model: margin sentence answering how a 3 cm tube absorbs everything from the Kenyan meal

Q2. A learner says: "The small intestine absorbs all the nutrients from ugali, sukuma wiki and nyama choma because it is a very wide tube." The small intestine is actually only about 3 cm wide. Which statement gives the best structural explanation for how it absorbs so much?

- A. Its sphincters hold nutrients in the intestine for several days
- B. Its villi and microvilli (brush border) greatly increase the surface area for absorption**
- C. Its strong hydrochloric acid breaks food into particles small enough to pass through
- D. Its wall is made of very thick muscle that squeezes nutrients into the blood

Correct answer: B. Villi and microvilli multiply the absorptive surface area, so a 6 m tube absorbs as much as a far longer smooth tube would. The intestine is narrow, and hydrochloric acid is in the stomach, not the small intestine.

USE AFTER: DQB PHASE — DQB: GREEN cards for the question 'How does the stomach avoid digesting itself?'

Q3. On the Driving Question Board, a group posts a GREEN card answering: "How does the stomach avoid digesting itself, even though it makes hydrochloric acid and pepsin?" Which answer is correct and complete?

- A. The stomach wall is covered by a mucus lining, and pepsin is released as inactive pepsinogen**
- B. Rugae fold away from the food so that acid never touches the wall
- C. The stomach wall contains no protein, so pepsin cannot harm it
- D. Pepsin works only in the small intestine, so the stomach never contains it

Correct answer: A. Mucus protects the stomach lining, and pepsin is released as inactive pepsinogen, so it only becomes active in the stomach's acid conditions. The stomach wall does contain protein, and pepsin acts in the stomach.

USE AFTER: EXPLAIN PHASE — Explain: annotating the stomach interior (rugae) on the Annotated Diagram Template

Q4. When a learner eats a large meal of ugali and nyama choma, the stomach stretches and churns the food. Which structure of the stomach wall lets it expand and increases contact with the food during churning?

- A. Villi
- B. Goblet cells
- C. Rugae**
- D. Microvilli

Correct answer: C. Rugae are folds of the stomach wall that unfold to let the stomach stretch and increase the surface in contact with food for mechanical and chemical digestion. Villi and microvilli are in the small intestine, and goblet cells make mucus.

USE AFTER: EXPLAIN PHASE — Explain: labelling the cardiac and pyloric sphincters on the simplified alimentary canal diagram

Q5. The cardiac sphincter is found where the oesophagus joins the stomach. What is its main role in digestion?

- A. It makes pepsinogen for protein digestion
- B. It absorbs glucose from the stomach into the blood
- C. It closes to stop stomach acid and food from flowing back up into the oesophagus (acid reflux)**
- D. It controls how quickly chyme leaves the stomach for the small intestine

Correct answer: C. The cardiac sphincter acts as a valve at the oesophagus-stomach junction and prevents acid reflux. Controlling the release of chyme is the job of the pyloric sphincter.

USE AFTER: EXPLAIN PHASE — Explain: whole-class discussion question about the pyloric sphincter opening too quickly

Q6. During the class discussion the teacher asks: "What would happen if the pyloric sphincter opened too quickly and released half-digested chyme into the small intestine before enzymes had finished working?" Which answer shows the correct role of the pyloric sphincter?

- A. It makes the stomach wall fold into rugae
- B. It keeps food in the oesophagus until the stomach is empty
- C. It produces mucus that stops the stomach digesting itself
- D. It controls the release of chyme from the stomach into the small intestine, so digestion is not rushed and can finish**

Correct answer: D. The pyloric sphincter sits at the stomach-duodenum junction and releases chyme in controlled amounts, giving enzymes time to work. If it opened too quickly, digestion would be incomplete.

USE AFTER: END OF LESSON — End of lesson: Gallery Walk prompt 'trace the exact path glucose from the ugali starch takes into the bloodstream'

Q7. Ugali starch has been digested to glucose by amylase and maltase. Which path correctly shows how this glucose gets into the blood in the small intestine?

- A. It is absorbed by the rugae of the small intestine into the blood
- B. It crosses the microvilli and epithelial cells of a villus into the capillary network inside the villus**
- C. It stays in the middle of the intestine and is carried out with the waste
- D. It passes through the pyloric sphincter directly into the blood

Correct answer: B. Glucose is absorbed across the microvilli (brush border) and epithelial cells of the villus, then enters the capillary network inside the villus and is carried away in the bloodstream. Sphincters control movement of food and do not absorb, and rugae are in the stomach.